

# Computational Strategies for the Digitization of Oral-Centric Gur Languages: A Comprehensive Framework for Dagbani Language Modeling and Synthetic Speech

The global landscape of Natural Language Processing (NLP) has arrived at a critical juncture where the technological divide between high-resource languages and low-resource, oral-centric languages threatens to marginalize significant populations from the benefits of the artificial intelligence revolution.<sup>1</sup> Dagbani, or Dagbanli, a member of the Mabia (Mole-Dagbani) subgroup of the Gur language family, serves as a quintessential case study for this challenge.<sup>4</sup> Spoken by over three million people in northern Ghana and parts of Togo, Dagbani is the primary lingua franca of its region, yet its digital presence remains disproportionately low compared to its cultural and demographic weight.<sup>4</sup> The inherent difficulty in building a Large Language Model (LLM) for Dagbani lies not only in the scarcity of digitized text but also in the fundamental nature of the language as an oral tradition.<sup>3</sup> For such communities, a text-based interface is insufficient; the architectural solution must encompass a multi-modal, speech-to-speech pipeline where Automatic Speech Recognition (ASR) and Text-to-Speech (TTS) are not peripheral additions but core pillars of the system.<sup>9</sup> This report provides an exhaustive analysis of the linguistic, technical, and sociotechnical requirements for constructing a representative, high-fidelity language model for Dagbani, prioritizing voice-first accessibility.

## Linguistic Architecture and Phonological Constraints

The development of effective NLP tools begins with a rigorous mapping of the target language's phonological and morphological structures, as these dictate the constraints of tokenization and prosodic modeling.<sup>4</sup> Dagbani presents unique computational hurdles characterized by a complex tonal system, a rich vowel inventory, and an agglutinative morphological structure.<sup>4</sup>

### The Tonal Landscape and Prosodic Modeling

Dagbani is a tonal language where pitch is utilized to distinguish lexical and grammatical meanings.<sup>4</sup> It operates on a two-level tone system—High (H) and Low (L)—augmented by the phenomenon of downstep, which is a lowering effect occurring between sequences of the same phonemic tone.<sup>5</sup> The semantic weight of tone is illustrated by minimal pairs such as *gballi*, which signifies 'grave' when produced with a High-High (H-H) pattern, but 'zana mat' when

produced with a High-Low (H-L) pattern.<sup>5</sup> For a language model to be truly representative, and for a TTS system to sound natural, the architecture must account for these tonal melodies.<sup>15</sup> In Dagbani nouns, only a subset of possible tonal patterns is prominent, with patterns like H-H, H-H-H, H-L, L-H, and L-L-H accounting for approximately 92% of usage.<sup>12</sup> These patterns often spread leftward from the right word boundary, suggesting that prosodic modeling should focus on the word-final transitions to capture authentic cadence.<sup>12</sup> Furthermore, Dagbani exhibits tonal polarity and dissimilation, particularly in the concatenation of high tones, which necessitates a tone-aware acoustic model in both ASR and TTS components.<sup>17</sup>

## Phonemic Inventory and Graphemic Representation

The vowel system of Dagbani is particularly dense, consisting of eleven phonemic vowels, categorized into six short and five long variations.<sup>5</sup> This density is further complicated by allophonic variations based on tongue-root advancement, which affects the realization of vowels like [i] ~ [ɪ] and [u] ~ [ʊ].<sup>5</sup>

| Vowel Type          | Front      | Central           | Back       |
|---------------------|------------|-------------------|------------|
| Short Vowels        | /i/, /e/   | /ɨ/ (or /ə/), /a/ | /u/, /o/   |
| Long Vowels         | /i:/, /e:/ | /a:/              | /u:/, /o:/ |
| Allophonic Variants | [ɪ], [ɛ]   | [ə]               | [ʊ], [ɔ]   |

The consonantal framework includes several digraphs and complex articulations such as labial-velar stops (/ɡb/, /kp/, /ŋm/) and palatal nasals (/ɲ/).<sup>4</sup> The presence of these digraphs requires specialized tokenization strategies, as standard character-level models might incorrectly treat them as discrete units, thereby losing the phonemic integrity of the word.<sup>4</sup> Furthermore, nasals in the coda position are tone-bearing units (TBU), meaning they bear both length and tone, making the mora the fundamental unit of prosody in Dagbani.<sup>13</sup>

## Morphological Complexity and Tokenization Challenges

As an agglutinative language, Dagbani forms words by attaching various prefixes and suffixes to roots, particularly within its system of six nominal classes.<sup>4</sup> This morphological richness results in a high degree of lexical variation, which poses a challenge for traditional word-level language models that may suffer from severe out-of-vocabulary (OOV) issues.<sup>2</sup> The solution lies in sub-word tokenization methods like Byte-Pair Encoding (BPE), which can decompose complex Dagbani words into their constituent morphemes, allowing the model to generalize across rare word forms.<sup>20</sup>

# Data Elicitation Strategies in Oral-Centric Environments

The primary obstacle to Dagbani language modeling is the "data desert" common to low-resource languages.<sup>1</sup> Because the language is predominantly oral, the available digitized text is often unrepresentative of natural, conversational speech.<sup>1</sup>

## The Image-Prompted Elicitation Paradigm

To build a model that resonates with the native speaker's experience, data collection must shift from text-based prompts to visual stimuli.<sup>23</sup> The UGSpeechData project and the WAXAL initiative have pioneered this approach in West Africa.<sup>23</sup> By presenting participants with culturally relevant images—covering themes such as market transactions, agricultural practices, and healthcare scenarios—researchers can elicit spontaneous, natural speech that contains the authentic prosody and disfluencies of a living language.<sup>23</sup> A significant effort in Ghana resulted in a 5,000-hour speech corpus, with 1,000 hours dedicated to Dagbani.<sup>24</sup> Each language corpus within this dataset includes audio descriptions of 1,000 culturally grounded images recorded via a mobile application.<sup>24</sup> Approximately 10% of this data has been meticulously transcribed by linguistic experts, providing the "golden" labeled data necessary for training high-accuracy ASR models.<sup>24</sup>

## Crowdsourcing and Community Knowledge Management

The revitalization of Dagbani in the digital space has been spearheaded by grassroots organizations like the Dagbani Wikimedians User Group.<sup>6</sup> This group has leveraged platforms like Wikipedia and Wikidata to create a structured repository of the language.<sup>4</sup> The Dagbanli dictionary on Wikidata is a landmark project, utilizing a lexicographical data model that breaks words into lexemes, senses, and forms.<sup>4</sup> This structured data is freely accessible and serves as a vital resource for training both NLU (Natural Language Understanding) and NLG (Natural Language Generation) components.<sup>4</sup> Furthermore, the integration of Mozilla Common Voice provides a scalable mechanism for ongoing data collection.<sup>35</sup> In Tamale, workshops have trained volunteers to contribute and validate Dagbani sentences, ensuring that the resulting open-source voice datasets reflect a wide range of dialects and accents, such as the Western Tomosili and Eastern Nayahali varieties.<sup>5</sup>

| Data Source  | Type                   | Utility in Dagbani NLP                                                    |
|--------------|------------------------|---------------------------------------------------------------------------|
| UGSpeechData | Audio (Image-Prompted) | High-fidelity ASR/TTS training; spontaneous speech patterns <sup>24</sup> |

|                             |                      |                                                                     |
|-----------------------------|----------------------|---------------------------------------------------------------------|
| <b>WAXAL Corpus</b>         | Transcribed Speech   | Cross-lingual transfer; phonetically balanced scripts <sup>23</sup> |
| <b>Wikidata</b>             | Lexicographical Data | Structured semantic mapping; dictionary integration <sup>4</sup>    |
| <b>Mozilla Common Voice</b> | Crowdsourced Audio   | Longitudinal dataset expansion; accent diversity <sup>35</sup>      |
| <b>Webonary / SIL</b>       | Dictionaries/Texts   | Orthographical grounding; lexical reference <sup>39</sup>           |

# Architectural Frameworks for Voice-First Language Assistants

Given the oral nature of Dagbani, the optimal approach is a cascaded speech-to-speech translation or interaction system.<sup>9</sup> This architecture consists of three interconnected modules: an ASR front-end, a central LLM cognitive engine, and a TTS back-end.<sup>9</sup>

## Automatic Speech Recognition (ASR): Breaking the Transcription Bottleneck

The ASR module is the most critical interface for an oral culture, as it converts the user's spoken intent into a machine-readable format.<sup>9</sup> For Dagbani, the "transcription bottleneck"—the fact that transcribing one minute of audio requires nearly an hour of human effort—has historically hindered progress.<sup>43</sup> To overcome this, contemporary researchers employ transfer learning from large-scale multilingual models like OpenAI's Whisper.<sup>19</sup>

While Whisper was not natively trained on Dagbani, its "weakly supervised" transformer architecture allows it to be fine-tuned on relatively small amounts of Dagbani data to achieve impressive Word Error Rates (WER).<sup>19</sup> Fine-tuning on 10-30 hours of high-quality transcribed audio can reduce error rates significantly, moving the technology from a research curiosity to a functional tool.<sup>19</sup> For morphologically rich languages like Dagbani, it is essential to monitor Character Error Rate (CER) alongside WER, as a single morphemic error can invalidate a word in WER while preserving its core characters and meaning.<sup>19</sup>

## Large Language Modeling and the DagBERT Implementation

The central LLM serves as the "brain" of the system. For Dagbani, Ghana NLP has introduced DagBERT, a model based on the BERT architecture specifically fine-tuned for Dagbani language

modeling.<sup>49</sup> Due to the lack of massive text corpora, developers typically use a two-step transfer learning process:

- 1. **Multilingual Pre-training:** Utilizing models like mBERT or XLM-RoBERTa that have been trained on over 100 languages, including several African languages.<sup>20</sup>
- 2. **Language-Specific Fine-tuning:** Training the model on the available Dagbani text from Wikipedia, religious texts (JW300), and crowdsourced dictionaries.<sup>20</sup>

Techniques like "back-translation"—translating English text to Dagbani and then back to English—can create synthetic parallel data that significantly improves the model's understanding of complex syntax and domain-specific vocabulary.<sup>2</sup> For lightweight deployment, researchers often use DistilBERT or RoBERTa architectures, which maintain high performance while having significantly fewer parameters, making them suitable for mobile hardware.<sup>20</sup>

### Text-to-Speech (TTS): The Pillar of Oral Accessibility

For a community whose culture is predominantly oral, TTS is the most important output mechanism, enabling the system to "speak" back to the user.<sup>42</sup> High-quality TTS requires clean, high-fidelity audio recordings from a single speaker to build a consistent synthetic voice.<sup>15</sup> The WAXAL-TTS dataset provides over 565 hours of such audio across various African languages, ensuring professional-grade acoustics through the use of custom-built studio

boxes.<sup>23</sup> For tonal languages, the TTS system must accurately model the  $F_0$  (fundamental frequency) contour to ensure that synthesized speech is not only intelligible but also conveys the correct lexical meaning.<sup>15</sup> Modern end-to-end architectures like VITS (Variational Inference with adversarial learning for end-to-end TTS) have shown great promise for low-resource languages, particularly when initialized from a related language model through cross-lingual transfer learning.<sup>53</sup>

### Sociotechnical Deployment: Infrastructure and Ethics

Building the model is only half the battle; ensuring its usability in Northern Ghana requires addressing the unique infrastructural and ethical landscape of the region.<sup>7</sup>

#### Connectivity and the "Offline-First" Mandate

In Ghana, while mobile connectivity is high (110% of the population has a cellular connection), internet penetration is approximately 69.9%, and many rural Dagbani speakers remain "unconnected" or face high data costs.<sup>56</sup> This creates a "usage gap" where advanced AI tools are theoretically available but practically inaccessible.<sup>57</sup>

| Infrastructure Metric | Ghana 2025 Statistic | Implications for AI Deployment |
|-----------------------|----------------------|--------------------------------|
|                       |                      |                                |

|                                    |                                       |                                                                      |
|------------------------------------|---------------------------------------|----------------------------------------------------------------------|
| <b>Mobile Internet Penetration</b> | 69.9%                                 | Significant population remains offline <sup>56</sup>                 |
| <b>4G/5G Network Coverage</b>      | Transitioning (MTN \$1.1B Investment) | Latency and reliability issues in rural areas <sup>60</sup>          |
| <b>Mobile Connection Type</b>      | Predominantly 3G/4G                   | High-bandwidth cloud AI may be prohibitively expensive <sup>60</sup> |
| <b>Regional Literacy Rates</b>     | Highly variable (Northern Region)     | Voice-first interfaces are essential for parity <sup>1</sup>         |

To solve this, the system must prioritize "offline-first" or "edge" AI.<sup>4</sup> This involves using model quantization and pruning to allow ASR, NMT, and TTS engines to run locally on a smartphone without needing an internet connection.<sup>9</sup> Frameworks like Vosk for ASR and Argos Translate for NMT enable real-time, on-device translation that is both secure and reliable in resource-constrained environments.<sup>9</sup>

## Ethics, Digital Humanism, and Cultural Sovereignty

The digitization of an oral culture is not a neutral act; it involves significant ethical considerations regarding cultural authenticity and community sovereignty.<sup>7</sup> The Masakhane Research Foundation, a grassroots organization, advocates for the "Ubuntu" philosophy—the idea that AI research should be for Africans, by Africans.<sup>63</sup>

Key ethical protocols include:

- **Data Sovereignty:** Native speakers should retain ownership of their voices and stories, preventing "extractive" research practices.<sup>8</sup>
- **Participatory Co-Creation:** Linguists, native speakers, and community elders should be involved in every stage of model development to ensure that the AI reflects the lived communication norms of the Dagbani people.<sup>7</sup>
- **Bias Mitigation:** Care must be taken to include a balanced representation of genders and dialects to prevent the AI from favoring urban or male speech patterns.<sup>31</sup>

## High-Impact Use Cases for the Dagbani Language Assistant

The ultimate goal of this technology is to improve quality of life and foster digital inclusion.<sup>42</sup>

1. **Precision Agriculture:** An AI app that can diagnose crop diseases from images and provide advice in spoken Dagbani, empowering rural farmers who may not read or write English.<sup>42</sup>
2. **Healthcare Communication:** Facilitating communication between health workers and patients in remote clinics, where language barriers can lead to diagnostic errors.<sup>1</sup>
3. **Educational Tools:** Enabling children and adults to learn and interact with educational

content in their native language, bridging the gap between L1 (first language) instruction and English.<sup>1</sup>

4. **Preservation of Oral Literature:** Digitizing and revitalizing folktales, proverbs, and oral histories that are at risk of being lost as the younger generation shifts toward global digital media.<sup>70</sup>

## Implementation Roadmap for the Dagbani LLM with TTS

Approaching the Dagbani language model project requires a phased strategy that builds on existing open-source frameworks and community datasets.<sup>3</sup>

### Phase 1: Foundation and Data Aggregation

The first step is to consolidate the available linguistic resources.<sup>3</sup> Developers should begin with the 1,000-hour Dagbani speech corpus from the UGSpeechData project and the transcribed subsets.<sup>24</sup> Simultaneously, the Lexemes from the Wikidata Dagbanli project should be used to build a robust sub-word tokenizer.<sup>4</sup>

### Phase 2: Training the Cascaded Speech-to-Speech System

The technical implementation should utilize state-of-the-art transfer learning.<sup>45</sup>

- **ASR:** Fine-tune the Whisper-small or Whisper-medium model using the transcribed Dagbani audio.<sup>19</sup> Implement pitch-aware features to better capture the tonal nuances.<sup>16</sup>
- **LLM:** Fine-tune DagBERT or a similar transformer model on the Dagbani Wikipedia and Common Voice text data.<sup>49</sup> Use back-translation to augment the training set with technical and professional terminology.<sup>2</sup>
- **TTS:** Train a VITS-based model using the high-fidelity recordings from the WAXAL dataset.<sup>23</sup> Focus on multi-speaker modeling to provide a choice of voices for the end-user.<sup>15</sup>

### Phase 3: Mobile-First Optimization and Deployment

Given the infrastructure in Northern Ghana, the final product must be optimized for local execution.<sup>9</sup>

- **Quantization:** Convert the models to INT8 or FP16 formats to reduce their memory footprint.<sup>9</sup>
- **Integration:** Develop an Android application (as Android is the dominant platform in the region) that integrates the ASR, LLM, and TTS modules.<sup>24</sup>
- **User Feedback Loop:** Implement a feature similar to the Khaya app, where users can submit corrections to improve the technology over time.<sup>76</sup>

## The Convergence of Orality and Artificial Intelligence

The challenge of building a language model for Dagbani is representative of the broader global struggle to preserve linguistic diversity in the digital age.<sup>2</sup> For the Dagbamba people, language is more than a tool for communication; it is a "reservoir of cultural wealth" and a "living archive of wisdom".<sup>3</sup> By adopting a voice-first, community-driven approach, developers can ensure that this cultural heritage is not only preserved but actively revitalized through AI.<sup>7</sup> The technical blueprint—combining image-prompted speech elicitation, transfer learning for ASR and TTS, and offline-optimized mobile deployment—provides a practical and scalable path to linguistic equity.<sup>9</sup> As Northern Ghana enters a new era of digital transformation, the integration of Dagbani into the digital ecosystem will stand as a testament to the power of AI to bridge the gap between ancient traditions and future innovations.<sup>42</sup>

*(Continuing with deep technical analysis of the LLM components...)*

## **Sub-word Embedding and Agglutinative Generalization**

In the Dagbani context, the vocabulary size is deceptive. While a dictionary might list 30,000 roots, the actual number of possible word forms is orders of magnitude higher due to the agglutinative nature of the language.<sup>4</sup> Traditional tokenizers like WordPiece, which are designed for English, often split Dagbani words into meaningless fragments.<sup>20</sup> The application of Byte-Pair Encoding (BPE), as used in the RoBERTa-based models like BAKO (BERT with Akan Knowledge Only), is a superior alternative.<sup>20</sup> BPE starts with individual characters and iteratively merges the most frequent pairs, which in Dagbani allows the model to "learn" common suffixes like *-li*, *-ga*, or *-gu* as distinct tokens.<sup>4</sup> This ensures that the model can understand the relationship between *wab-gu* (elephant) and *wab-ri* (elephants) by identifying the shared root *wab-*.<sup>12</sup>

Furthermore, the research emphasizes the use of "decoupled embeddings." Models like RemBERT and XLM-V reallocate parameters from the embedding layer into the Transformer layers, which has been shown to yield significant performance gains for African and indigenous languages where the vocabulary bottleneck is a primary constraint.<sup>50</sup> By using a larger, inclusive vocabulary of up to 1 million tokens, these models can better represent the morphological diversity of Dagbani without sacrificing computational efficiency.<sup>50</sup>

## **Modeling Tone in Synthetic Speech: Acoustic and Linguistic Strategies**

The synthesis of Dagbani speech requires a dual-track approach to tone.<sup>16</sup> On the acoustic

side, the model must predict the  $F_0$  trajectory from the input text.<sup>15</sup> On the linguistic side, the text must be diacritically marked to provide the necessary cues for the model.<sup>18</sup>

A critical finding in the study of Yoruba ASR—which applies directly to Dagbani—is that explicit tone marking (diacritics) significantly influences the accuracy of the system.<sup>18</sup> However, in common Dagbani orthography, tone is rarely marked.<sup>14</sup> This creates a discrepancy between the written form and the spoken reality. To resolve this, researchers propose a "phonetic front-end" for the TTS system that uses a rule-based or machine-learning model to automatically insert tone markers into the input text before it is passed to the acoustic model.<sup>15</sup>

For the acoustic model, the use of "adversarial" training (as seen in MelGAN architectures) can help the model generate higher-frequency details that are essential for distinguishing between subtle tonal shifts.<sup>80</sup> These models use multiple discriminators to ensure that the generated audio is not only clear but also matches the prosodic "fingerprint" of a native Dagbani speaker.<sup>30</sup>

## **The Role of LLMs in Speech-to-Text Enhancement**

The LLM is not just for generating responses; it also plays a vital role in refining ASR outputs.<sup>45</sup> A "language model re-scoring" pass can take the top candidates from the ASR system and use the DagBERT model to determine which one is most grammatically and contextually likely in Dagbani.<sup>49</sup> This is particularly helpful in noisy environments (e.g., a busy market in Tamale), where the acoustic signal might be ambiguous.<sup>81</sup>

Moreover, the integration of Voice Activity Detection (VAD) within the African Whisper framework helps to minimize "hallucinations"—where the model generates text during silences or background noise—a common failure mode in low-resource ASR.<sup>45</sup> By aligning the transcription outputs with Wav2vec models, the system can provide word-level timestamps, which are essential for creating synchronized educational content and subtitled media in Dagbani.<sup>45</sup>

## **Sociopolitical Context: AI as a Tool for Language Revitalization**

The mission of building a Dagbani language model is situated within the broader context of "Digital Humanism," which prioritizes human values and cultural diversity in technological innovation.<sup>67</sup> The "Sankofa" framework—an African philosophy that encourages looking to the past to inform the future—is often cited as a guiding principle for these projects.<sup>72</sup> Digitizing oral tradition is not seen as an attempt to replace elders or the storytelling tradition, but rather as a way to "reclaim the fire" of indigenous storytelling in an era dominated by smartphones and global media.<sup>72</sup>

The Ghanaian government's "AI Ready Ghana" program and the National AI Strategy are designed to equip the youth with the skills to build these local language technologies.<sup>68</sup> By fostering a "robust AI community ecosystem," the strategy aims to turn Ghana into a trailblazer for AI leadership in Africa, using languages like Dagbani as a testing ground for inclusive, responsible AI.<sup>68</sup>

*(Continued elaboration... focusing on the specific findings of the WAXAL-ASR demographic analysis, the impact of code-switching between Dagbani and English on model performance, and the detailed comparative analysis of BLEU vs. CHRF scores for West African translation models...)*

The metrics for evaluation in this domain are evolving.<sup>1</sup> While BLEU (Bilingual Evaluation Understudy) remains the standard for translation, it is often criticized for failing to account for the morphological complexity of African languages.<sup>1</sup> Newer metrics like CHRF (Character n-gram F-score) and TER (Translation Edit Rate) are increasingly used to provide a more nuanced view of how well a model captures the structure of a language like Dagbani.<sup>1</sup> For ASR,

the transition toward CER is particularly vital for tonal languages where a single character change (representing a tone shift) can completely change the meaning of a sentence.<sup>19</sup> As the research and implementation of the Dagbani language model progress, the focus must remain on the three-fold challenge of data scarcity, linguistic complexity, and infrastructural constraints.<sup>3</sup> Through the collaborative efforts of the Dagbani community, the academic rigor of Ghana NLP, and the ethical frameworks provided by Masakhane, the goal of a voice-first, culturally authentic Dagbani language assistant is well within reach.<sup>4</sup> This technology will not only bridge the digital divide but will ensure that one of West Africa's most vibrant oral traditions continues to thrive in the age of artificial intelligence.

The final architectural recommendation is a **Community-Stewarded Multimodal Hub**. This hub would serve as the centralized repository for the DagBERT model, the WAXAL-derived TTS voices, and the UGSpeechData-based ASR engines.<sup>23</sup> By hosting these resources on open platforms like Hugging Face and GitHub, the project ensures that any developer—whether in Tamale, Accra, or the global diaspora—can build applications that speak, understand, and preserve the Dagbani language.<sup>49</sup> This is the essence of digital inclusion: turning a "digital silence" into a vibrant, multi-vocal presence that empowers three million people to be heard in their own tongue.

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